# Efficient Brain MRI Classification with Validation Guided Transfer Learning

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**Abstract**

This project reproduces a selected Xception baseline from Disci et al. (2025) and investigates a smaller classifier for four-class brain MRI classification. We use the pinned 7,023-image dataset and distinguish the original folder-based protocol from an audited split that excludes known exact training–test overlap. The selected baseline reproduction reaches 93.67% clean test accuracy, compared with the paper's 95.27%. Under the audited protocol, the initial MobileNetV2 replacement fails the original performance-preservation hypothesis. Correcting its ImageNet input normalization improves macro-F1 to 0.8062. We then compare a longer-training control with a compact model combining Global Average Pooling, frozen-backbone warm-up and low-rate fine-tuning with frozen BatchNorm statistics. The compact model reaches 92.68% test accuracy and 0.9256 macro-F1 with 2,422,468 parameters, 90.33% fewer than the historical audited Xception reference (0.9176 macro-F1). The final exploratory efficiency targets are met in this run. Single-seed results, a previously examined test set and missing patient identifiers limit generalization claims. Established techniques are evaluated; no new algorithm or statistical superiority is claimed.

**Keywords:** brain MRI; tumor classification; transfer learning; MobileNetV2; Xception; reproducibility.

## 1 Introduction

Brain MRI classification provides a practical setting for studying transfer learning when labeled images and computational resources are limited. Our task assigns individual MRI slices to glioma, meningioma, no tumor or pituitary tumor. A useful capstone result should report predictive performance together with model size and enough experimental evidence to reproduce the comparison.

Disci et al. evaluate six pretrained convolutional networks on a public MRI collection and identify Xception as their best reported model [1]. We select that paper and its Xception model to establish one baseline. The scope is the selected Xception model. MobileNetV2 is our proposed smaller replacement within this baseline-and-extension study.

Our work has three practical contributions. First, we implement and document an approximate selected-model replication. Second, we audit exact image overlap and preserve both favorable and unfavorable experiment outcomes. Third, we test a compact classification head and validation-guided adaptation, with exported checkpoints and a saved-model prediction demonstration. These contributions concern experimental application and reproducibility, not algorithmic novelty.

## 2 Selected Paper and Motivation

The base paper uses transfer learning, preprocessing and augmentation on 7,023 images [1]. Its Table 8 reports Xception test accuracy of 95.27% and macro-F1 of 0.9491. The headline 98.73% weighted

**Table 1:** Original dataset version 1 folder counts.

| Partition | Glioma | Meningioma | No tumor | Pituitary | Total |
|---|---|---|---|---|---|
| Training | 1,321 | 1,339 | 1,595 | 1,457 | 5,712 |
| Testing | 300 | 306 | 405 | 300 | 1,311 |

accuracy is consistent with a sample-weighted combination of training and testing accuracy, as inferred from its tables; we therefore use the Table 8 test score for comparison.

The paper identifies class-recall challenges, explainability, dataset diversity and clinical validation as areas requiring further investigation. Our selected extension examines the accuracy–parameter tradeoff and training stability. MobileNetV2 uses inverted residuals and linear bottlenecks to support efficient feature extraction [2]. Its architecture motivates testing a smaller alternative, without assuming that its pretrained features will transfer successfully to this MRI dataset.

The first smaller-model attempt leaves a substantial performance gap. This motivates two questions: whether a longer validation-guided training policy helps, and whether a smaller head with staged fine-tuning improves the tradeoff. The warm-up and low-rate adaptation follow established transfer-learning practice, including careful treatment of BatchNorm [3].

# 3 Dataset and Evaluation Protocols

## 3.1 Pinned dataset

We use version 1 of Masoud Nickparvar's Brain Tumor MRI Dataset [4]. It contains 7,023 images, matching the base paper. Version 2 contains 7,200 images and is excluded. Table 1 gives the original folder counts. The download URL, archive SHA256 and ordered image manifests are recorded.

## 3.2 Overlap audit and splits

Hashing decoded image pixels identifies 263 training images identical to test images, 207 redundant images within training and 30 redundant images within testing. No exact duplicate has conflicting labels. A perceptual-hash audit also identifies 1,254 unresolved near-match candidate pairs; these are flagged candidates, not verified patient matches.

We retain two distinct protocols. The *paper-based* protocol keeps the original folders and evaluates clean images and one fixed seeded transformed test view, reflecting the source paper's test-augmentation description. The *audited* protocol removes exact training–test overlap and redundant training copies, then reserves 15% of the remaining training pool for validation. It yields 4,518 training, 798 validation and 1,311 test images. Training alone receives augmentation in the audited protocol.

Internal test duplicates remain. Reliable patient identifiers are unavailable, so neither exact deduplication nor this split establishes patient independence. All final experiments use the same ordered split fingerprint. Differences between paper-based and audited results involve several protocol changes and cannot isolate the effect of leakage removal.

# 4 Methodology

## 4.1 Shared preprocessing

Each slice is read in grayscale, smoothed with a Gaussian kernel of size 5, thresholded at 45, and cropped to the bounding rectangle of the largest foreground contour. The blurred grayscale crop is resized to

**Table 2:** Predetermined control and proposed training policies.

| Setting | A: training control | B: compact proposal |
|---|---|---|
| Aggregation | Existing Flatten | Global Average Pooling |
| Total epoch budget | Up to 20 | 5 warm-up plus up to 15 fine-tuning |
| Warm-up | None | Backbone frozen; head rate $10^{-4}$ |
| Fine-tuning | All layers at $10^{-4}$ | Non-BatchNorm layers at $10^{-5}$ |
| BatchNorm statistics | Updated | Frozen throughout |
| Checkpoint | Minimum validation loss | Minimum validation loss across both stages |

$128 \times 128$, replicated into three channels and scaled to $[0,1]$. The training pipeline applies brightness and contrast factors in $[0.8,1.2]$. Foreground cropping succeeds on all 7,023 images without a contour fallback. It is not a validated tumor-segmentation procedure.

Kernel size, threshold, bounding-rectangle cropping, seed 42 and full backbone trainability are disclosed implementation assumptions where the paper does not provide a complete specification. These assumptions limit exact replication. For normalization follow-ups, the model converts $[0,1]$ pixels to the documented MobileNetV2 ImageNet range $[-1,1]$ [5]. The operation is saved inside the model, so inference does not normalize inputs twice.

## 4.2 Baseline and original replacement

The selected ImageNet backbone is followed by Flatten, Dropout(0.3), a 128-unit ReLU dense layer, Dropout(0.2), and a four-unit Softmax output. The initial experiments fine-tune the complete backbone using Adam with learning rate $10^{-4}$, batch size 20, sparse categorical cross-entropy and five epochs. They evaluate the final-epoch checkpoint, preserving the replication policy rather than selecting a checkpoint after viewing test scores.

The original hypothesis, H1, is that under identical audited settings, MobileNetV2 will achieve macro-F1 no more than 0.02 below Xception while using fewer total parameters. Both conditions are required. This original result remains separate from all later exploratory experiments.

## 4.3 Compact head and staged adaptation

The final proposal replaces Flatten with Global Average Pooling (GAP). For each channel $c$ of a feature tensor with height $H$, width $W$ and $C$ channels, GAP computes

$$g_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F_{ijc}, \qquad c = 1, \ldots, C. \tag{1}$$

Flatten preserves all $HWC$ entries; GAP produces $C$ values, reducing the input size of the unchanged dense classifier. At our input resolution this reduces the complete MobileNetV2 model from 4,880,068 to 2,422,468 parameters. It does not provide a tumor mask or prove that the model attends to a lesion.

We define exploratory hypothesis H2 as improvement over normalized five-epoch MobileNetV2 macro-F1 while using fewer parameters. The secondary target is a macro-F1 gap of at most 0.02 from the saved audited Xception reference, also with fewer parameters. Table 2 specifies the two new experiments. Both start afresh from ImageNet weights and use identical audited data and preprocessing.

B completes five warm-up epochs, then begins fine-tuning from the final warm-up weights. We recompile with a fresh Adam optimizer at the stage boundary. BatchNorm layers remain frozen during fine-tuning. Head dropout remains active during training. Fine-tuning stops after four consecutive epochs without lower validation loss; its stopping counter resets at the stage boundary. The best checkpoint is retained globally across both stages, including warm-up.

**Table 3:** Paper-based Xception replication, separate from audited experiments.

| Evaluation | Test accuracy (%) | Macro-F1 |
|---|---|---|
| Published Table 8 [1] | 95.27 | 0.9491 |
| Our clean test view | 93.67 | 0.9354 |
| Our fixed transformed view | 93.29 | 0.9315 |

A tests a combined longer-budget and validation-selection policy. A versus B tests a combined pooling, learning schedule and BatchNorm intervention. These comparisons do not identify the individual causal effect of each component.

# 5 Experimental Setup

## 5.1 Metrics and evidence

For four classes, we report ordinary test accuracy and macro-F1,

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N}, \qquad F_{1,\text{macro}} = \frac{1}{4}\sum_{c=1}^{4} \frac{2\text{TP}_c}{2\text{TP}_c + \text{FP}_c + \text{FN}_c}. \tag{2}$$

Per-class recall, $\text{TP}_c/(\text{TP}_c + \text{FN}_c)$, exposes uneven errors. Weighted F1, balanced accuracy and ROC-AUC are retained in the machine- readable evidence. Parameter reduction against reference $R$ is $100(1 - P/P_R)\%$, using total model parameters rather than trainable parameters.

We independently recompute metrics and confusion matrices from 1,311 ordered saved predictions per experiment. Histories, manifests, configuration identity, checkpoint SHA256 values and source-code matches are checked before accepting the returned Colab runs. No new accuracy is inferred from a small demo sample.

## 5.2 Runtime and checkpoint selection

The new runs use the free Colab GPU workflow selected for a T4, Python 3.12.13, TensorFlow 2.18.1, Keras 3.8.0, NumPy 2.0.2 and batch size 20. Recorded core packages, platform and device descriptions match between A, B and the normalization run. Historical Xception has a different recorded runtime platform. Seed 42 and deterministic data ordering do not guarantee bitwise GPU equivalence.

A completes 19 epochs, selects epoch 15 and has minimum validation loss 0.2617. B completes 20 epochs, selects epoch 17 and has minimum validation loss 0.0937. The lower validation loss also selects B for demonstration, independently of test ranking. Test data never enters training or checkpoint-selection callbacks. Because earlier experiments have already examined this test set, the final comparison is exploratory rather than independent confirmatory validation.

# 6 Results

## 6.1 Selected baseline reproduction

Table 3 reports the paper-based baseline. Our clean Xception accuracy is 1.60 percentage points below the published test accuracy. The transformed view is 1.98 points below it. This is an approximate selected-model replication, not an exact numerical reproduction of the entire paper.

**Table 4:** Audited test results. Historical Xception uses different input scaling, training budget and checkpoint policy; this is not a matched architecture comparison.

| Model or experiment | Accuracy (%) | Macro-F1 | Parameters |
|---|---|---|---|
| Historical Xception | 91.76 | 0.9176 | 25,056,428 |
| Original MobileNetV2 | 74.29 | 0.7372 | 4,880,068 |
| Normalized MobileNetV2 | 81.92 | 0.8062 | 4,880,068 |
| A: longer-training control | 88.71 | 0.8841 | 4,880,068 |
| **B: compact proposal** | **92.68** | **0.9256** | **2,422,468** |

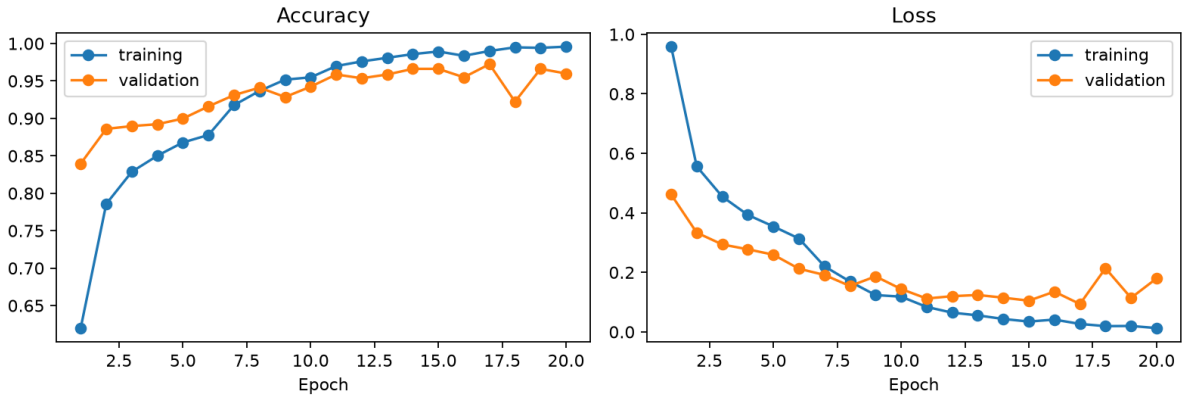



**Figure 1:** Proposed model training and validation curves over 20 epochs. The checkpoint is chosen by minimum validation loss at epoch 17, rather than the final epoch or the test score. The first five epochs train the head only.

## 6.2 Original and normalization experiments

Under the audited five-epoch protocol, original MobileNetV2 reaches 74.29% accuracy and 0.7372 macro-F1, compared with Xception's 91.76% and 0.9176. The 0.1804 macro-F1 gap exceeds the allowed 0.02 margin. H1 is therefore unsupported despite an 80.52% parameter reduction.

Changing only the configured input normalization improves MobileNetV2 to 81.92% accuracy and 0.8062 macro-F1, an observed gain of 7.63 accuracy points and 0.0690 macro-F1. Its 0.1114 gap to historical Xception still misses the target. The earlier and normalized runs have different recorded runtime platforms, so normalization is not established as the sole cause of the gain.

## 6.3 Final performance and efficiency

Table 4 gives the complete audited progression. B improves on the normalized five-epoch model by 10.76 accuracy points and 0.1194 macro-F1 while using 50.36% fewer parameters. Relative to A, B gains 3.97 accuracy points and 0.0415 macro-F1. H2's observed improvement-and-size criterion is met.

Relative to the saved audited Xception, B is 0.92 accuracy points and 0.0080 macro-F1 higher, with 90.33% fewer parameters. Its observed gap satisfies the secondary 0.02-margin target. This does not establish statistical superiority or non-inferiority. B also uses a different protocol from the paper's published 95.27% result; it must not be presented as outperforming that result.

Training takes 643.1 seconds (10.72 minutes) for A and 382.3 seconds (6.37 minutes) for B, including validation and checkpoint writes. Warmed batch-1 median inference times are 5.29 and 5.19 milliseconds, respectively, from 20 synchronized calls excluding image I/O and preprocessing. These are descriptive single-run timings. Historical Xception trains for 239.2 seconds over five epochs, so B does not train faster than that saved reference.

**Table 5:** Class recall on the same audited test set.

| Class | A recall (%) | B recall (%) |
|---|---|---|
| Glioma | 91.00 | 93.67 |
| Meningioma | 84.97 | 82.35 |
| No tumor | 96.30 | 96.05 |
| Pituitary | 80.00 | 97.67 |

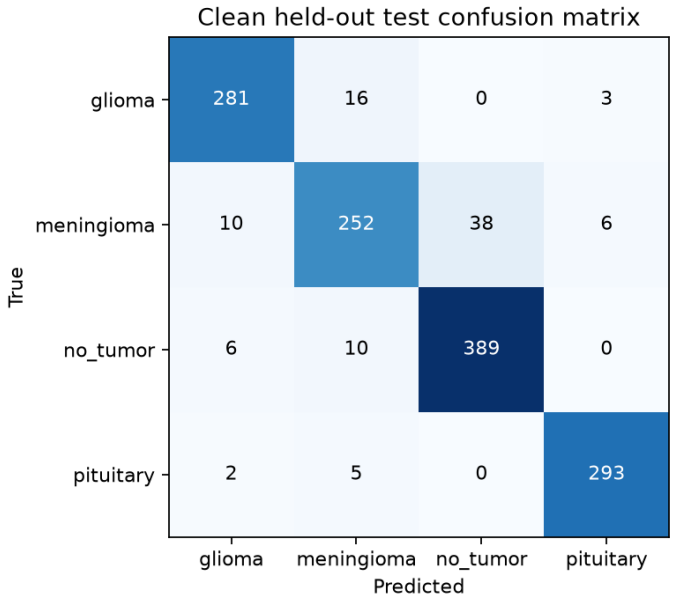



**Figure 2:** Proposed model confusion matrix from all 1,311 saved Colab test predictions. Rows are dataset labels and columns are predictions.

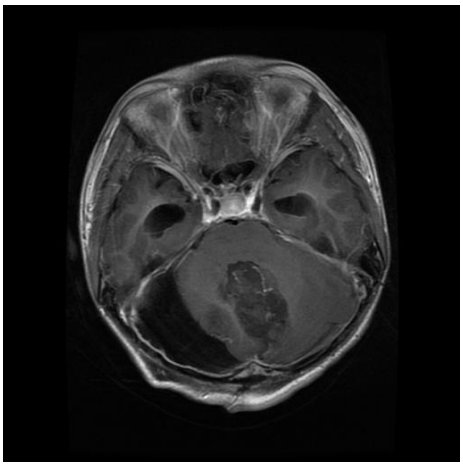
### 6.4 Class-level error analysis

Table 5 and Fig. 2 show that improvement is not uniform. Pituitary recall rises from 61.67% in normalized five-epoch MobileNetV2 to 97.67% in B. However, B's meningioma and no-tumor recall are slightly lower than A's. In particular, 38 of 306 meningioma slices are predicted as no tumor, motivating further error analysis rather than an accuracy-only conclusion.

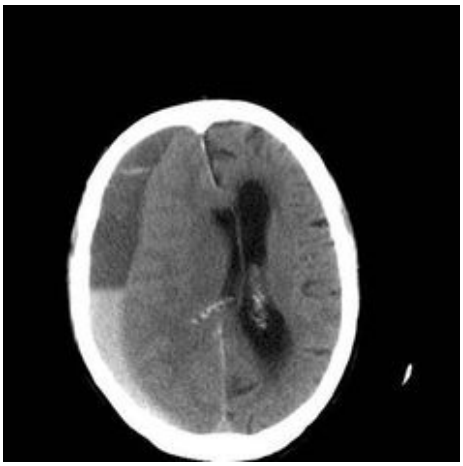
## 7 Saved Model Demonstration and Reproducibility

Both accepted checkpoints load on the Mac CPU with the expected parameter counts; all 52 BatchNorm layers are frozen in B's saved model. Four deterministic first-per-class test images have verified decoded-pixel hashes. B predicts three of four correctly, while A predicts two of four. All eight local predicted classes agree with the corresponding saved Colab classes, including errors. Maximum absolute CPU–GPU score differences are 0.0145 for B and 0.0538 for A; their cause is unestablished. We do not claim strict probability equivalence or separately reproduced full-test CPU accuracy.

Figure 3 uses the same clean glioma image shown in the Colab demonstration and a recorded no-tumor error. The examples are the first test file in each of these classes, not an estimate of class performance. Both examples have verified dataset provenance. Original images are shown with their dataset and predicted labels; full-test prediction records provide the quantitative evaluation.



(a) Dataset label: glioma
Prediction: glioma



(b) Dataset label: no tumor
Prediction: meningioma

**Figure 3:** Correct and incorrect predictions from the proposed checkpoint, consistent between the checked Mac CPU and saved Colab classes. Images are from dataset version 1 [4]. These examples do not establish diagnosis, tumor localization or calibrated medical probabilities.

The repository contains the original executed Colab notebook, the baseline reproduction notebook, the normalization notebook and the final training-extension notebook. The latter displays verified saved results and model inference with `RUN_TRAINING=False`. The shared pipeline exposes simple `train`, `analyze` and `predict` commands. Forty-four automated tests cover baseline behavior, staged training, checkpoint selection, BatchNorm handling, evidence validation and resume checks. Synthetic test runs are not reported as MRI research results. Full model exports are retained locally; weights and the full raw dataset are excluded from Git.

# 8 Discussion and Limitations

The final experiment provides a positive exploratory accuracy–size result and an informative longer-training control. A's improvement demonstrates that the initial five-epoch normalized model was not the strongest result obtainable with that architecture and a validation-guided training policy. B further improves macro-F1 with a smaller head and staged adaptation. Because the changes are bundled, the result cannot isolate whether pooling, lower-rate adaptation, BatchNorm handling or their interaction is responsible.

Several limitations remain. Only one seed is evaluated. Imaging modality and clinical labels are taken from the supplied dataset and are not independently verified. The test set has been examined across project stages, so it is not an untouched confirmation set. Exact overlap removal leaves unresolved perceptual near-matches and internal test duplicates; missing patient identifiers prevent patient-level guarantees. Historical Xception uses a different input-scaling convention, training budget, selection policy and recorded runtime. Foreground cropping lacks segmentation annotations, and no external clinical validation is performed. The observed margin is not a statistical non-inferiority test. The method combines established components rather than introducing a new learning algorithm.

Further work should repeat seeds, separately ablate pooling and fine-tuning choices, inspect persistent meningioma errors, and obtain patient-level or independent external validation. These steps would test the stability and scope of the present finding.

# 9 Conclusion

We establish an approximate selected Xception replication and preserve an initial unsupported smaller-model hypothesis. Normalization improves the first MobileNetV2 result but does not preserve the target performance. A subsequent compact, staged adaptation reaches 92.68% accuracy and 0.9256 macro-F1, with 90.33% fewer parameters than the saved audited Xception reference. The final exploratory improvement and efficiency targets are met in this run. The executed notebooks, checked prediction evidence, saved checkpoints and reported errors support a reproducible demonstration while keeping the remaining validation limits explicit.

# References


[1] Rukiye Disci, Fatih Gurcan, and Ahmet Soylu. Advanced brain tumor classification in MR images using transfer learning and pre-trained deep CNN models. *Cancers*, 17(1):121, 2025. `https://doi.org/10.3390/cancers17010121`.

[2] Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. MobileNetV2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 4510–4520, 2018. `https://arxiv.org/abs/1801.04381`.

[3] François Chollet. Transfer learning and fine-tuning. Keras Developer Guides, 2023. `https://keras.io/guides/transfer_learning/`. Accessed 2 November 2026.

[4] Masoud Nickparvar. Brain tumor MRI dataset, version 1. Kaggle. Pinned version used in this project. `https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset/versions/1`. Accessed 2 November 2026.

[5] TensorFlow. MobileNetV2 input preprocessing. TensorFlow API Documentation. `https://www.tensorflow.org/api_docs/python/tf/keras/applications/mobilenet_v2/preprocess_input`. Accessed 2 November 2026.


# A Reproduction Information

The code and notebooks are available at `https://github.com/visheshrao17/ComputerVision_Prj/tree/feat/final-evaluation-normalization`. The implementation and imported experiment evidence are recorded in commit `bfbd165`. The LaTeX source, bibliography and figure assets accompany this report and can be compiled in Overleaf with `main.tex` as the main document. No GPU is required to read the stored notebook outputs.

The pinned dataset archive SHA256 is

`048b7413ce05bb52627046cfa5cde097b12e865aa80f6110959f547fd6fe4ef1`

Experiment settings are in `configs/training_control.yaml` and `configs/training_proposed.yaml`; compact evidence is under `results/training_extension`. The reported split fingerprint is

`7d26f705e2895e05fbebdcb5487f2fa011ebc27ee71d2d04a2a93598d50af571`

Both example images are illustrative report assets; the complete raw dataset and model checkpoints are distributed separately from Git.